

TITLE: Study of PCI and SCSI.

Batch: A2

Roll No.: 16010122034

Experiment / assignment / tutorial No. 01

AIM: To Study and learn PCI and SCSI

Expected OUTCOME of Experiment : (Mention CO/CO's attained here)

CO1- Describe and define the structure of a computer with buses structure and detail working of the arithmetic logic unit and its sub modules

Books/ Journals/ Websites referred:

1. <https://www.techopedia.com/definition/8815/peripheral-component-interconnect-bus-pci-bus>
2. <https://www.techopedia.com/definition/331/small-computer-system-interface-scsi>
3. https://www.csun.edu/~edaasic/roosta/BUS_Structures.pdf
4. W.Stallings William "Computer Organization and Architecture: Designing for Performance", Pearson Prentice Hall Publication, 7thEdition. C.

Pre Lab/ Prior Concepts:

Microcomputer buses which communicate with peripheral devices or a memory location through communication lines called buses.

The major parts of microcomputers are central processing unit (CPU), memory, and input and output unit. To connect these parts together through three sets of parallel lines, called buses. These three buses are Address bus, data bus, and Control bus.

Address Bus:

The address bus consists of 16, 20, 24, or more parallel signal lines, through which the CPU sends out the address of the memory location. This memory location is used for writing to or reading from. The number of memory locations depends on 2 to the power N address lines. Example, a CPU with 16 address lines can address 2¹⁶ or 65,536 memory locations. When the CPU reads data from or writes data to a port. The port address is also sent out on the address bus. This is unidirectional. This means that the CPU can send data to a memory location or I/O ports.

Data Bus:

The data bus consists of 8, 16, 32 or more parallel signal lines. The data bus lines are bidirectional. This means that the CPU can read data from memory or from an I/O port as well as send data to a memory location or to a I/O port. In a system, many output devices are connected to the data bus, but only one device at a time will be enabled to the output.

Control Bus:

The control bus consists of 4-10 parallel signal lines. The CPU sends out signals on the control bus to enable the outputs of addressed memory devices or port devices. Typically control bus signals are memory read, memory write, I/O read and I/O write. To read data from a memory location, the CPU sends out the address of the desired data on the address bus and then sends out a memory read signal on the control bus. The memory read signal enables the addressed memory device to output the data onto the data bus where it is read by the CPU.

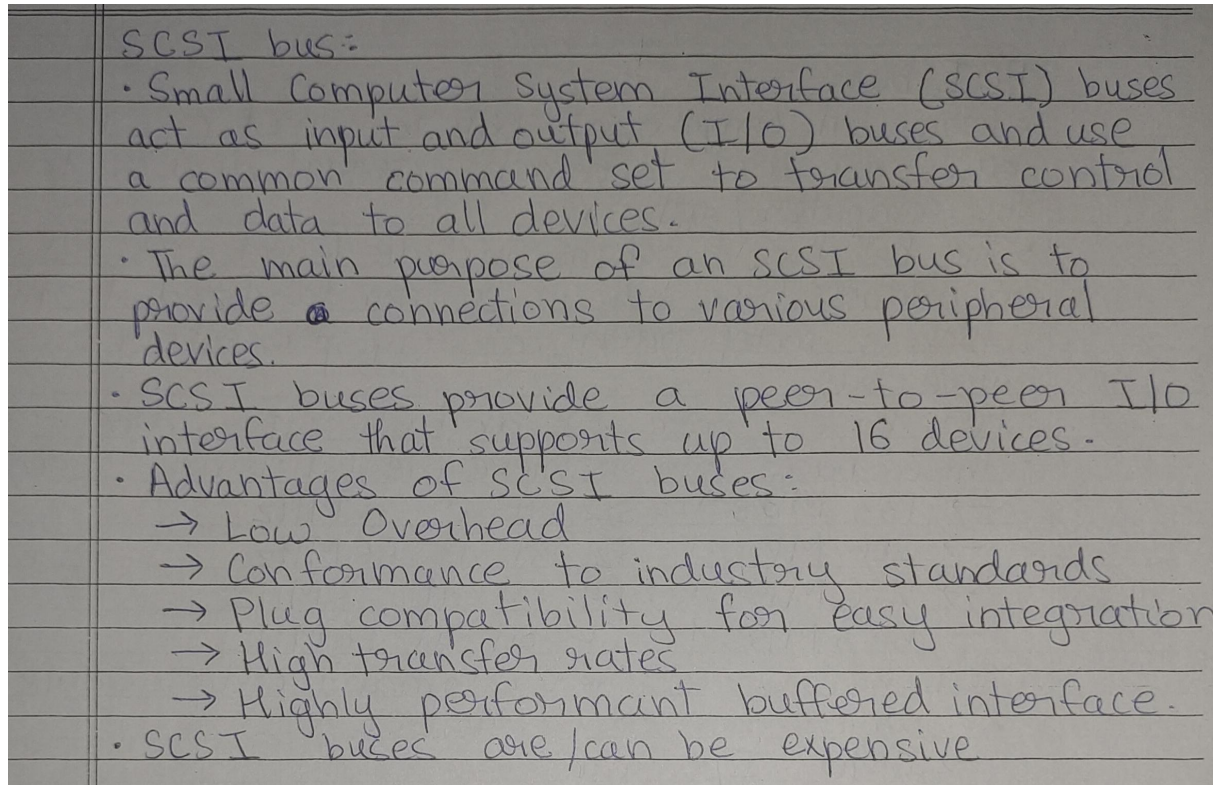
PCI Bus:

PCI bus:

- Peripheral Component Interconnect (PCI) buses are used to attach external devices to the computer/system.
- The PCI bus functions include functions of an address bus but in a standardized form - which is independent of any processor's native bus.
- PCI buses have 32 and 64 bit widths
- These buses can achieve the following speeds:
 - 133 MB s^{-1} : 32 bit at 33 MHz
 - 266 MB s^{-1} : 32 bit at 66 MHz
 - 266 MB s^{-1} : 64 bit at 33 MHz
 - 533 MB s^{-1} : 64 bit at 66 MHz
- PCI has 12 assigned command codes. Some of which are as follows:
 - 0010 : I/O Read
 - 0011 : I/O Write
 - 0110 : Memory Read
 - 0111 : Memory Write
- Advantages of PCI buses:
 - It supports linear burst mode data transfer
 - It offers a large bandwidth and high speeds
 - It supports plug and play
 - Also supports bus mastering
- Disadvantages of PCI buses:
 - It is a parallel bus - which means that the available bandwidth is shared by all connected devices - and, so, it has bandwidth limitations
 - It does not support advanced power management.



SCSI bus:



Post Lab Descriptive Questions

Q1 . Differentiate between PCI and SCSI Bus

SCSI buses are used to interact with I/O devices (example: external hard drives) while PCIs are used for interacting with various peripheral devices.

Q2. List two applications each of PCI and SCSI Bus

Applications of PCI are:

1. Network cards
2. Sound cards

Applications of SCSI:

1. Hard drives
2. Tape drivers

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